

Module 05: SQL Clauses, Aggregate Functions, and Groupings

SQL Clauses Order of Execution & Aggregation Pipeline

Written Syntax Order vs True Physical Engine Execution Order, GROUP BY & HAVING

Physical SQL Execution Order (Engine)

1. FROM & JOIN
Identifies tables and calculates Cartesian product / Joins.

2. WHERE (Row Filtering)
Filters individual tuples BEFORE grouping. No aggregate functions allowed!

3. GROUP BY (Bucket Creation)
Groups rows having identical values in specified columns into summary buckets.

4. HAVING (Group Filtering)
Filters grouped buckets AFTER aggregation (e.g. HAVING COUNT(*) > 5).

5. SELECT & Expressions
Evaluates selected columns, expressions, aliases and DISTINCT keyword.

6. ORDER BY (Sorting)
Sorts final resulting records ASC / DESC. Column aliases can now be used.

7. LIMIT / OFFSET: Top-N rows

Aggregates & WHERE vs HAVING

The 5 Core Aggregate Functions:

- COUNT(*): Counts total rows including NULLs.
- COUNT(col): Counts non-NULL entries in that column.
- SUM(col): Sums non-NULL numeric values.
- AVG(col): Computes arithmetic mean (ignores NULLs!).
- MIN(col) / MAX(col): Minimum / maximum value.

Golden Rule: All aggregates IGNORE NULLs (except COUNT(*)!)

WHERE vs HAVING (10-Marker Comparison):

- Level: WHERE operates on individual rows; HAVING on grouped buckets.
- Timing: WHERE executes BEFORE GROUP BY; HAVING executes AFTER.
- Aggregates: WHERE cannot use SUM/AVG; HAVING can use all aggregates.
- Can exist without GROUP BY: WHERE yes; HAVING acts as whole table filter.

```
SELECT Dept, AVG(Salary) FROM Employee
WHERE Salary > 30000 GROUP BY Dept
HAVING AVG(Salary) > 60000;
```

Execution Pipeline: The physical database engine executes FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY.

1. WHERE vs HAVING Comparison

Feature	WHERE	HAVING
Scope	Individual rows before grouping.	Grouped summary buckets after aggregation.
Aggregates	Forbidden (e.g. WHERE AVG(x) is invalid).	Allowed (e.g. HAVING COUNT(*) > 5).
Timing	Filters input tuples early.	Filters groups late in pipeline.

2. Aggregate Functions

All aggregate functions (SUM, AVG, MIN, MAX, COUNT(col)) strictly ignore NULL entries. Only COUNT(*) counts all rows including NULL records.